

Polymarket Scalping Bot Strategy Analysis

Executive Summary

This trader is operating a **high-frequency scalping bot** on Polymarket, executing approximately **10 trades per minute** across Bitcoin and Ethereum "Up or Down" hourly markets. The bot employs multiple strategies simultaneously to extract small profits from market inefficiencies.

Total Activity Analyzed: 3,556 trades over ~6 hours on November 5, 2025

Core Strategies Identified

1. Binary Arbitrage (Primary Strategy)

How it works: Exploits mispricing in binary markets where Up + Down should equal \$1.00

Evidence:

- Found multiple instances where both sides cost less than \$1.00
- Examples:
 - Buy Up @ 21¢ + Sell Down @ 78¢ = 99¢ → **1¢ profit per pair**
 - Buy Up @ 93¢ + Buy Down @ 5¢ = 98¢ → **2¢ profit per pair**
 - Buy Up @ 82¢ + Buy Down @ 17¢ = 99¢ → **1¢ profit per pair**

Why this works: Binary outcomes must sum to 100% probability. When market inefficiencies cause the total to be less than \$1.00, buying both guarantees profit when the market resolves.

2. Momentum Scalping

How it works: Buy positions and quickly flip them for 1-3¢ profit per share

Evidence from recent trades:

- Buy BTC Up @ 47¢ → Sell @ 48¢ = **+1¢/share** (62 shares = \$0.62 profit)
- Buy BTC Up @ 75¢ → Sell @ 90¢ = **+15¢/share** (16 shares = \$2.40 profit)
- Buy ETH Down @ 39¢ → Sell @ 60¢ = **+21¢/share** (22 shares = \$4.62 profit)
- Buy BTC Up @ 24¢ → Sell @ 30¢ = **+6¢/share** (10 shares = \$0.60 profit)

Holding Period: Typically 1-20 trades (seconds to minutes)

Strategy: Enter when price moves against the trend, exit when it reverts or momentum continues

3. Mean Reversion Trading

How it works: Buy extreme outcomes (very high or very low probabilities) expecting reversion to 50/50

Evidence:

- **82 trades** buying "Down" at >65¢ (betting on unlikely outcomes becoming more likely)
- **23 trades** buying "Up" at <35¢ (same logic)
- **23 trades** selling near 50¢ (taking profit when price normalizes)

Logic: Hourly crypto price movements are hard to predict. Extreme probabilities (80%+ or 20%-) often overcorrect, creating opportunities when they revert toward 50/50.

4. Market Making

How it works: Provide liquidity by simultaneously posting bids and offers, capturing the spread

Evidence:

- Negative spreads observed: Buying at 42.6¢, Selling at 37.5¢ (BTC 8AM market)
 - This appears backwards, but suggests:
 - Bot is earning **maker rebates** (Polymarket pays liquidity providers)
 - Bot is accumulating directional positions for later profit
 - Bot is providing liquidity for fee income, not just spread capture
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Trading Statistics

Volume & Frequency

Metric	Value
Total Trades	3,556
Buy Orders	2,268 (64%)
Sell Orders	1,286 (36%)
Trades per Minute	~10
Average Trade Size	\$19.40

Price Distribution

Side	Average Price	Range
Buys	60.5¢	3¢ - 99¢
Sells	55.9¢	2¢ - 99¢

Key Insight: Bot buys at higher average prices than it sells. This suggests it's either:

1. Earning liquidity rebates that offset the "loss"
2. Accumulating positions for market close
3. Making money on volume rather than per-trade margins

Market Focus

- **Bitcoin markets:** 2,660 trades (75%)
 - **Ethereum markets:** 829 trades (25%)
 - **Preferred markets:** Hourly "Up or Down" binary options
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Position Sizing Patterns

Standard Trade Sizes

The bot uses specific share quantities repeatedly:

- **10 shares** - Standard small position (~\$4-7 each)
- **37 shares** - Medium position (~\$15-22 each)
- **62 shares** - Large position (~\$25-40 each)
- **75 shares** - Very large position (~\$30-60 each)

Strategic Sizing

- **Arbitrage trades:** Matched pairs (same size on both sides)
- **High-confidence trades:** 100-387 shares when edge is clear
- **Odd lots:** 1-5 shares for position cleanup or testing

Risk Management

Key Observations:

1. **Time-limited exposure:** Only trades hourly markets, limiting downside
2. **Binary outcomes:** Maximum loss is capped at purchase price
3. **Diversification:** Spreads trades across multiple concurrent markets
4. **Rapid turnover:** Most positions held for minutes, reducing overnight risk

Potential Risks:

- **Accumulation risk:** More buys than sells suggests net long positions
- **API latency:** Bot needs fast execution; delays could turn profits to losses
- **Fee sensitivity:** Small 1-3¢ profits depend on low transaction costs
- **Market resolution:** Must exit before hourly close or rely on correct outcome

Estimated Profitability

Per-Trade Margins:

- **Arbitrage trades:** 1-2¢ per share (100% win rate)
- **Momentum scalps:** 1-15¢ per share (depends on exit timing)
- **Mean reversion:** 5-20¢ per share (longer holding period)

Volume-Based Profit Model:

Assuming conservative estimates:

- **3,556 trades** at average **20 shares** = **71,120 shares total**
- Average profit of **2¢ per share** = **\$1,422 gross profit**
- Over 6 hours = **\$237/hour** or **\$5,688/day** if sustained

Note: This excludes:

- Transaction fees (typically 2% on Polymarket)
 - Gas fees for blockchain transactions
 - Losing trades where price moved against the bot
 - Maker rebates (which could add 0.5-1% back)
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Technical Implementation Insights

Bot Architecture:

1. **WebSocket connection** for real-time price feeds
2. **Order book monitoring** to detect arbitrage opportunities
3. **Automated order placement** via Polymarket API
4. **Position tracking** across multiple markets simultaneously
5. **Risk limits** enforced by position sizing rules

Execution Speed:

- **Sub-second latency** required for arbitrage
- **Co-location advantages:** Likely running on cloud servers near Polymarket infrastructure
- **Parallel processing:** Handles multiple markets concurrently

Key Algorithms:



python

Pseudo-code for main strategies:

1. Arbitrage Scanner

```
if (up_price + down_price) < 1.00:  
    buy_both_sides()  
    profit = 1.00 - (up_price + down_price)
```

2. Momentum Scalper

```
if price_change > threshold:  
    enter_position()  
    if profit >= 2¢ or loss >= 5¢:  
        exit_position()
```

3. Mean Reversion

```
if price > 0.70 or price < 0.30:  
    enter_contrarian_position()  
    if price approaches 0.50:  
        exit_position()
```

Competitive Advantages

What Makes This Bot Effective:

1. **Speed:** Executes before human traders can react
2. **Consistency:** Never misses opportunities due to fatigue/emotion
3. **Multi-strategy:** Doesn't rely on single edge
4. **Volume:** Profits accumulate through thousands of small wins
5. **Automation:** Operates 24/7 without breaks

Market Inefficiencies Exploited:

- **Psychological biases:** Humans overreact to recent price moves
- **Liquidity gaps:** Bot fills orders when spreads widen
- **Probability errors:** Markets sometimes misprice binary outcomes
- **Speed advantage:** Arbitrage disappears in milliseconds

Conclusion

This is a **sophisticated, multi-strategy scalping bot** designed to extract small, consistent profits from Polymarket's binary markets. The core edge comes from:

1. **Binary arbitrage** (guaranteed profits when Up + Down < \$1.00)
2. **High-frequency momentum scalping** (1-3¢ per trade × thousands of trades)
3. **Mean reversion** (fading extreme probabilities)

4. **Market making** (earning liquidity rebates)

The bot's success depends on:

-  Extremely fast execution
-  Low transaction costs
-  Consistent market inefficiencies
-  High trading volume

Estimated daily profit: \$2,000-\$6,000 based on observed activity (before costs)

Risk level: Medium-Low (short holding periods, capped downside, diversified)

Recommendations for Competitors

If trying to replicate or compete with this strategy:

1. **Focus on speed:** Co-locate servers, optimize code for latency
 2. **Fee structure matters:** Ensure maker rebates exceed costs
 3. **Start small:** Test with 10-share positions before scaling
 4. **Monitor fill rates:** If <90% of orders fill, you're too slow
 5. **Diversify strategies:** Don't rely solely on arbitrage (it's rare)
 6. **Risk management:** Set hard stops to prevent runaway losses
 7. **Continuous optimization:** Markets adapt; your bot must too
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Analysis Date: November 5, 2025

Data Source: Polymarket user activity

Analysis Method: Pattern recognition from 3,556 trades